

How Far Off the Recorded Path Can a Gaussian-Splat Driving Simulator Be Trusted?

CS 566 Project Proposal (one document per team) · Kyle Sha (ID 9085525476) · Allen Xia · September 29, 2026

Main papers: [1] **Difix3D+** (CVPR 2025 Oral), *reimplemented* · [2] **FaithFusion** (CVPR 2026), *compared* · [3] **HUGSIM** (TPAMI 2026), *simulator used for testing*

1. Problem (Q1)

Self-driving software is increasingly tested in simulators built from real drives. The recorded video is turned into a **3D Gaussian splat** [13], a 3D scene made of millions of small coloured blobs that can be rendered from any camera position, so the simulated car can drive *differently* from the recording. The problem: **the splat is only accurate along the path the car actually drove**. When the simulated car changes lanes or the camera sits higher (e.g., a truck), the rendered images degrade, and a driving system tested on them may behave differently than it would in reality. We ask: **how far from the recorded path can we move before the simulator stops being trustworthy, and do current repair methods ("fixers") restore trust or just make images look nicer?** We define **trust** as agreement between what a lane detector and a car detector see in the rendered image and in the **real** photo from the same position.

2. Why it matters and why we care (Q2)

Simulation is how driving software gets tested at scale; if off-path renders are wrong, a simulator can pass unsafe software or fail safe software. Kyle leads controls on a student self-driving team (Wisconsin Autonomous) and works on vehicle simulation, so "when can we believe the simulator?" is a question we face directly. It is also squarely a vision problem: camera geometry, 3D reconstruction, image restoration and CNN-based perception.

3. Current state of the art (Q3)

- **Splat-based driving simulators:** HUGSIM [3] runs full driving policies in reconstructed scenes; the EUVS benchmark [7] shows that current methods break down under large viewpoint changes.
- **Diffusion "fixers":** Difix3D+ [1] cleans off-path renders with a one-step diffusion model and feeds them back into the 3D scene (reported $\sim 2\times$ FID improvement over prior methods); NVIDIA Fixer [4] applies the same recipe to driving data; FaithFusion [2] limits how much a fixer may overwrite, using a per-pixel confidence map.
- **Evaluation practice:** these methods are judged mostly by image-realism scores (FID, LPIPS) or by shifting cameras without any real image at the new position (PRG-Fusion [6]); a 2026 study [10] measures detector trust vs. offset but without fixers. SetDiff [11] and MV2 [9] use real off-path photos, but only with image-similarity metrics and without fixers. **No prior work compares fixers head to head on what they invent, against real off-path photos.**

4. Our approach: re-implementation plus new evaluation (Q4)

- **Re-implement:** Difix3D+ [1] (the fixer's training and its "progressive update" that feeds fixed images back into the 3D scene), written from the paper on the same pretrained diffusion model the authors used, validated against NVIDIA's official code and weights, then trained on our driving data.
- **New (our contribution):** (a) an **evaluation with real ground truth** (and training pairs from neighbouring real lanes, cf. ReCamDriving [12]), using Para-Lane [8], which scans the same roads in three parallel lanes (≈ 3.3 m and 6.3 m apart), and MV2 [9], a car and a scooter filming the same road at different camera heights; (b) **hallucination accounting:** invented or erased lane paint and fake vehicles, not just image similarity; (c) a **lightweight fixer trained from scratch** (a 15M-parameter U-Net, about $60\times$ smaller than Difix's diffusion backbone); (d) a **geometry-only road prior**, adapted from ViSE [15], that flattens the road using LiDAR and cannot invent content; (e) a **transfer test** on a different country and camera (EUVS/MARS [7], MV2 [9]).

5. Why existing approaches fall short, and why ours should help (Q5)

Existing fixers are diffusion models trained to produce *realistic* images, so they are rewarded for plausible detail even where it is wrong. Our pilot (Sep 28) confirms this: 3 m off-path, lane-detector agreement with the real photo drops from 0.98 to 0.07, and NVIDIA's released Difix improves realism scores much more than lane accuracy (lane agreement $0.07 \rightarrow 0.31$; lane-paint precision unchanged). Our hypotheses: **(H1)** training a fixer on real adjacent-lane pairs makes it more faithful, not just prettier (pilot on 2 held-out test roads: lane agreement $0.07 \rightarrow 0.71$, though it adds fake vehicles, 0.18 vs. 0.04 per image); **(H2)** a geometry-only prior improves lanes with zero hallucination; **(H3)** fixers trained on one city will not transfer, so trust must be measured per domain.

6. Evaluation and planned results (Q6)

- **Deviation axes:** lateral offset (main; real photos at 3.3 m / 6.3 m), camera height (main; real car ↔ scooter pairs), heading (secondary; exact image warp, seen regions only).
- **Metrics:** image similarity to the real photo (PSNR, LPIPS); lane-paint precision *and* recall (invented vs. erased paint); lane-detector (CLRNet [14]) and car-detector (YOLO11) agreement, following ReconDreamer's metrics [5]; fake-vehicle count; and, as a stretch goal, a driving policy's score inside HUGSIM [3].
- **Compared:** raw render, released Difix, NVIDIA Fixer, our re-implemented Difix, our lightweight U-Net, progressive update, FaithFusion weighting, road prior.
- **Planned results:** *Fig. 1* trust vs. lateral offset per method; *Fig. 2* trust vs. camera height; *Table 1* hallucination counts per method; *Table 2* cross-dataset transfer; *Table 3* our Difix vs. official code; side-by-side videos with invented content marked.

Component	In the paper	Our version
Difix fixer [1]	Diffusion fixer; official code and weights released	Re-implemented and retrained by us on driving pairs; checked against official
Progressive update [1]	Fixed images fed back into the 3D scene	Re-implemented
FaithFusion [2], NVIDIA Fixer [4]	Released code / weights	Run as baselines
Lightweight fixer; road prior [15]	—; road-surface model	New, trained from scratch; adapted and re-implemented

7. Timeline (Q6a)

Dates	Kyle	Allen
9/29–10/12	MV2 scenes and camera-height axis; more Para-Lane test roads	Evaluation harness for MV2 / EUVS; per-scene variance
10/13–10/29	Difix re-implementation validated vs. official code	Road prior on all scenes; hallucination analysis. Mid-term report 10/29
10/30–11/20	Transfer study; progressive update on all scenes	HUGSIM driving-policy tests (stretch)
11/21–12/10	Final figures and videos	Presentation (12/1–12/10) and webpage (12/10)

Risks: results rest on few test roads (we add scenes and report per-scene variance); older driving policies need a legacy software stack (we use the one that runs, LTF). Project webpage: ksha23.github.io/cs566-drive-sim.

References

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6. PRG-Fusion, 2026 · arxiv.org/abs/2609.06948
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10. From Concept to Capability (3DGS for AD), SafeComp 2026 · arxiv.org/abs/2605.01995
11. SetDiff, 2026 · arxiv.org/abs/2601.07540
12. ReCamDriving, ECCV 2026 · arxiv.org/abs/2512.03621
13. 3D Gaussian Splatting, SIGGRAPH 2023 · arxiv.org/abs/2308.04079
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